Azure Local Architecture Overview

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Agenda

- **Objective:** Review reference architecture for Azure Local (infrastructure-focused, workload-agnostic).
- **Scope:** Compute, storage, networking, identity, Azure integration security posture, update management, deployment workflow

Azure Local Basics

- **Definition:** On-premises/edge Azure-consistent HCI platform delivering VMs, AKS, and Arc-enabled services under Azure governance.
- **Control Plane:** Azure Arc projects on-prem resources into Azure Resource Manager for centralized management.
- **Scale:** From 1 to 16 (hyperconverged) or 64 (disaggregated) physical machines, validated OEM solutions.
- **Primary Use Cases:** Latency-sensitive edge processing, data sovereignty, standardized multi-site ops.

Reasons for Deployment

- **Data Sovereignty:** Keep data on-prem to satisfy regulation/compliance while retaining Azure governance & billing.
- **Latency:** Execute workloads close to data (manufacturing, retail, healthcare), minimizing round-trip times.
- **Resiliency:** Run HA VMs/AKS on multi-node clusters; survive node, network, or Azure control-plane disruptions.
- **Consistency:** Use Azure Portal/CLI/ARM/Bicep/Policy/Monitor uniformly across cloud & edge.
- **Security:** Enforce secure-by-default posture; integrate Defender for Cloud; maintain drift control on nodes.
- **At-Scale Ops:** Roll out repeatable deployments to many sites (retail chains, plants) with IaC + Arc.

Hardware Choices

Premier Solutions

Turnkey, specific appliance models

One-step automated update process

Single point of contact (OEM) with proactive 'call home'

Factory-imaged, ready for immediate deployment

Mission-critical workloads, AI workloads, and minimal IT intervention

Integrated Systems

Specific OEM product lines

Quarterly/periodic vendor update bundles

Single point of contact (OEM)

Pre-installed, but requires more setup config

Standard data center modernization and branch offices

Validated Nodes

Mix & match from validated component list

Manual (You track and apply patches)

Component-level or separate hardware/software support

DIY assembly and OS installation

Custom setups, test environments, or maximum component control

Building Blocks

- **Azure Local (HCI):** Hosts VMs, AKS, and Arc-enabled services with Storage Spaces Direct and validated hardware.
- **Azure Arc:** Extends Azure Resource Manager to on-prem, enabling RBAC, Policy, Monitor, Update Manager.
- **Azure Key Vault:** Stores secrets, certificates, keys for workloads and infra; integrates with Arc/VM/AKS operations.
- **Cloud Witness:** Provides quorum in 2-node clusters to maintain HA.
- **Update Manager:** Unifies OS/firmware/agent updates for nodes and guest OS patching for VMs.

Deployment Types

- **Hyperconverged**: 1-16 machines using hyperconverged storage
  - **Compute and storage** run on the **same physical nodes**
  - **S2D**, **simplified scaling**, uniform hardware, and support for **rack-aware clusters** (up to 4-4 nodes)
- **Disaggregated**: 1-64 machines using SAN storage
  - **Compute nodes** separate from **storage nodes**
  - **Large-scale**, **high-performance**, or **independent scaling** environments
- **Multi-rack**: pre-integrated racks with shared SAN storage, compute, and networking for large-scale deployments.
- **Disconnected**: no connection to Azure public (3+ nodes, extra capacity required)
- **Small form factor**: compact, single-node edge for container, K3s, AI, and IoT.
- **Low-capacity**: 1-3-node edge for lightweight, resource-constrained scenarios.
- **Microsoft 365 Local**: hosting Microsoft 365 Local workloads.

Hyperconverged Architecture

- Definition:
  - **Compute**, **storage**, and **networking** integrated on **each node**
- Benefits:
  - **Unified cluster management**
  - **High throughput** via **Storage Spaces Direct**
  - **Simplified cabling & deployment**
  - **Consistent performance** across nodes
- Requirements:
  - **Identical hardware configuration** across all nodes
  - **Local storage devices** used for S2D

Disaggregated Architecture

- Definition:
  - **Compute hosts**: Run VMs
  - **Storage hosts**: Provide a **separate storage fabric**
- Benefits:
  - **Independent scaling** of compute vs storage
  - Suitable for **large datasets** or **specialized workloads**
  - Supports **SAN-based storage** scenarios
- Requirements:
  - **Supported SAN solutions** (Fibre Channel only)
  - Consistent HBA zoning & LUN presentation

Identity Options for Azure Local

- Active Directory (AD)
  - Provides centralized authentication, OU structure, Group Policy control
  - Requires domain join, LCM deployment user, AD preparation
- Local Identity with Azure Key Vault
  - AD-less deployment model
  - Uses local administrator accounts
  - Secrets stored in Azure Key Vault (BitLocker keys, credentials, etc.)
- Relevance:
  - Both fully supported; AD is preferred for enterprises

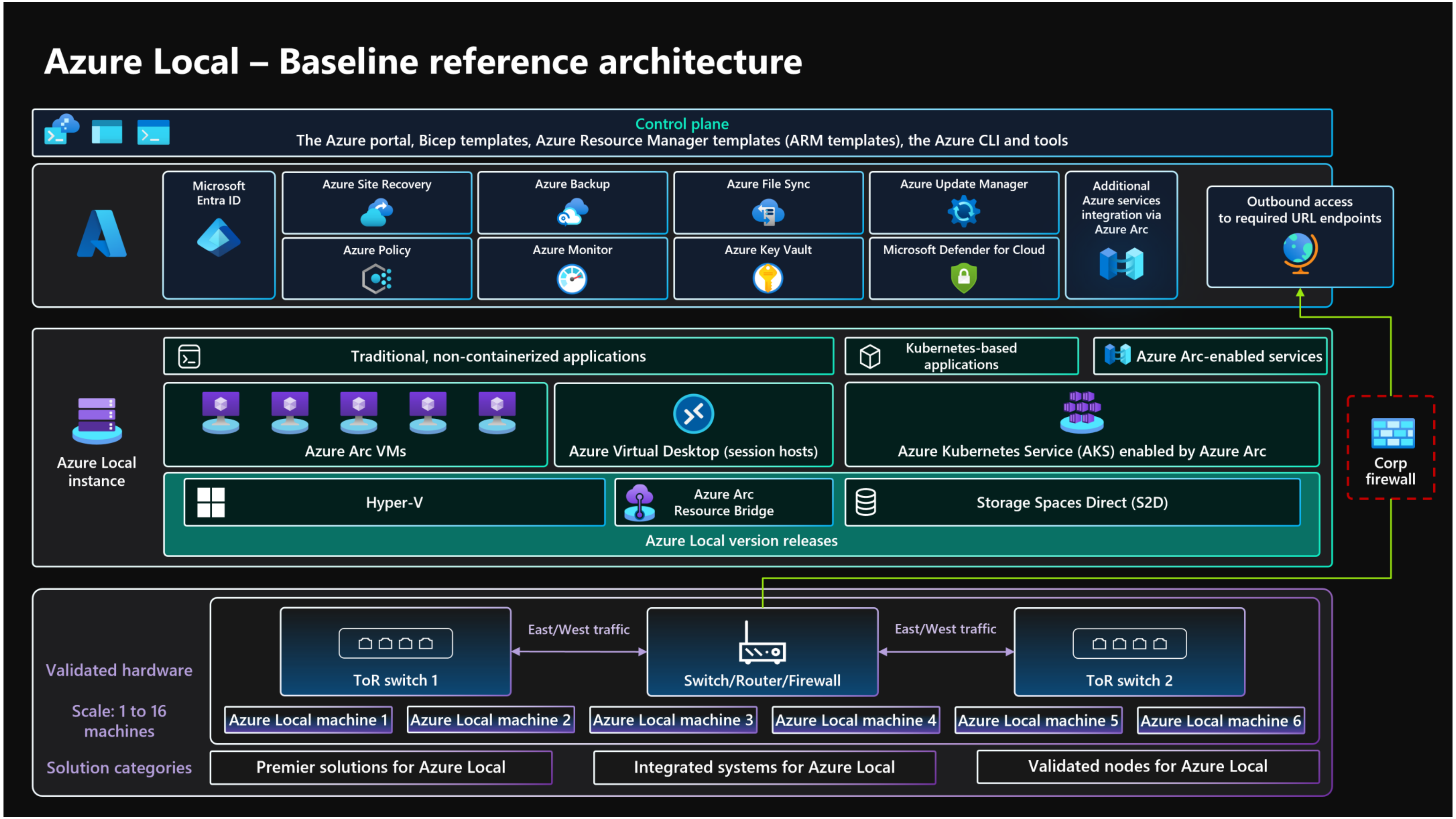
Supported Workloads

- **Virtual Machines (Arc-enabled):** Traditional apps/services; deploy via Portal/CLI/ARM/Bicep; managed with extensions.
- **AKS on Azure Local:** Container orchestration for microservices; supports Container Apps and Functions (via extension).
- **Azure Virtual Desktop (AVD):** On-prem session hosts orchestrated by Azure control plane.
- **Azure Edge RAG (Preview)**: All-in-one, on-premises pipeline automating local data ingestion, vector search, and context retrieval..
- **Foundry Local on Azure Local (Preview)**: Shared, on-premises inference engine hosting and serving open-source AI models locally.
- **Azure IoT Operations**: IoT on AKS to manage and process IoT data at the edge.
- **Azure Video Indexer**: Insights from videos by using video and audio AI models

Reference Architecture (Hyperconverged)

- **Multi-Node Cluster:** 2–16 nodes, dual ToR switches; management/compute converged NICs + dedicated RDMA storage NICs.
- **Azure Connectivity:** Arc Resource Bridge; Outbound egress to required endpoints; no TLS inspection allowed.
- **Azure Services:** Monitor/Insights, Policy, Defender for Cloud, Update Manager, Backup, Azure Site Recovery.
- **Storage:** S2D with ReFS + CSV volumes; three-way mirroring recommended at ≥3 nodes.
- **Networking:** VLANs for storage (non-routable) and compute/management; MLAG between ToR switches.

Reference Architecture (Hyperconverged)



Sizing & Capacity

- **Workload Inventory:** Define VM/Pod counts, vCPU, memory, GPU needs, storage capacity/IOPS.
- **Resiliency Targets:** Choose N+1 (general) vs N+2 (mission-critical).
- **Node Count:** Map requirements to 1–16 nodes; consider switchless storage limits (≤4 nodes) where applicable.
- **Performance Considerations:** Latency/throughput expectations per app (DB, analytics, AKS, AVD).
- **Growth Plan:** Model future scale (compute, memory, storage) for 3–5 years.
- **Tooling:** Use Azure Local Sizing Tool + OEM sizing aids.

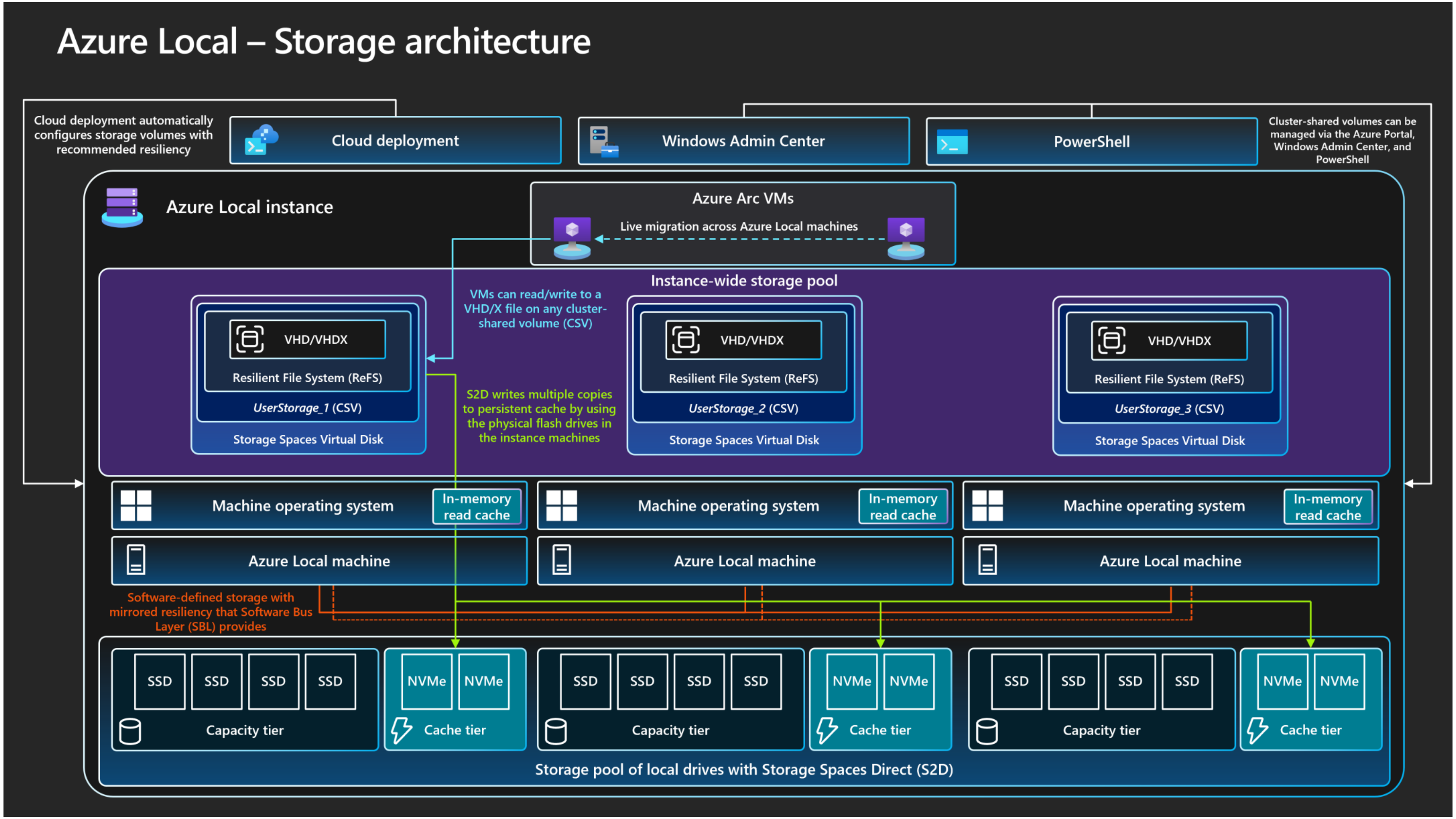
Resiliency Design Choices (N+1 vs N+2)

- **N+1:** Reserve one node worth of headroom for updates/failures; suitable for many business workloads.
- **N+2:** Reserve two nodes worth of headroom; recommended for mission-critical services.
- **Operational Scenarios:** Live migration during updates; simultaneous maintenance + unexpected failure.
- **Storage Resiliency:** Three-way mirroring at ≥3 nodes; balances performance, fault tolerance.
- **Risk Posture:** Higher cost, but lower outage probability.

Storage Architecture (Hyperconverged)

- **Drive Types: NVMe/SSD** (performance), **HDD** (capacity), **PMem** (advanced use cases).
- **Caching:** Server-side persistent cache (read/write) sized to application working set.
- **File System: ReFS** on **CSV** for resilience and performance.
- **Transport: SMB Direct over RDMA** (RoCE v2/iWARP) with **PFC** for lossless storage classes.
- **Volumes:** Balance resiliency (mirroring/parity), capacity efficiency, performance per workload profile.

Storage Architecture (Hyperconverged)



Storage Architecture Patterns (Hyperconverged)

- **All-Flash Design: NVMe or all-SSD** for latency-sensitive workloads; maximize throughput; no cache-tier reservations.
- **Hybrid Design: NVMe/SSD cache + HDD capacity**; maximize usable TB; watch cache **working set** limits.
- **Mixed Tiers:** Combine NVMe + SSD capacities; understand cost/performance trade-offs.
- **Volume Planning:** Choose **mirroring** where performance matters most; consider parity for archive/backup tiers.
- **CSV In-Memory Read Cache:** Improves **Hyper-V VM** performance (unbuffered access).

Validating Storage Performance

- **DiskSpd:** Generate synthetic read/write workloads; measure latency, throughput, IOPS with parameter tuning.
- **VMFleet:** Simulate VM-backed load across nodes; validate S2D subsystem under realistic stress.
- **Baseline Establishment:** Capture pre-production benchmarks; compare to OEM specs; document results.
- **Alerting:** Use Insights to track storage health metrics; set threshold-based alerts.
- **Operational Tuning:** Adjust cache and block sizes per workload.

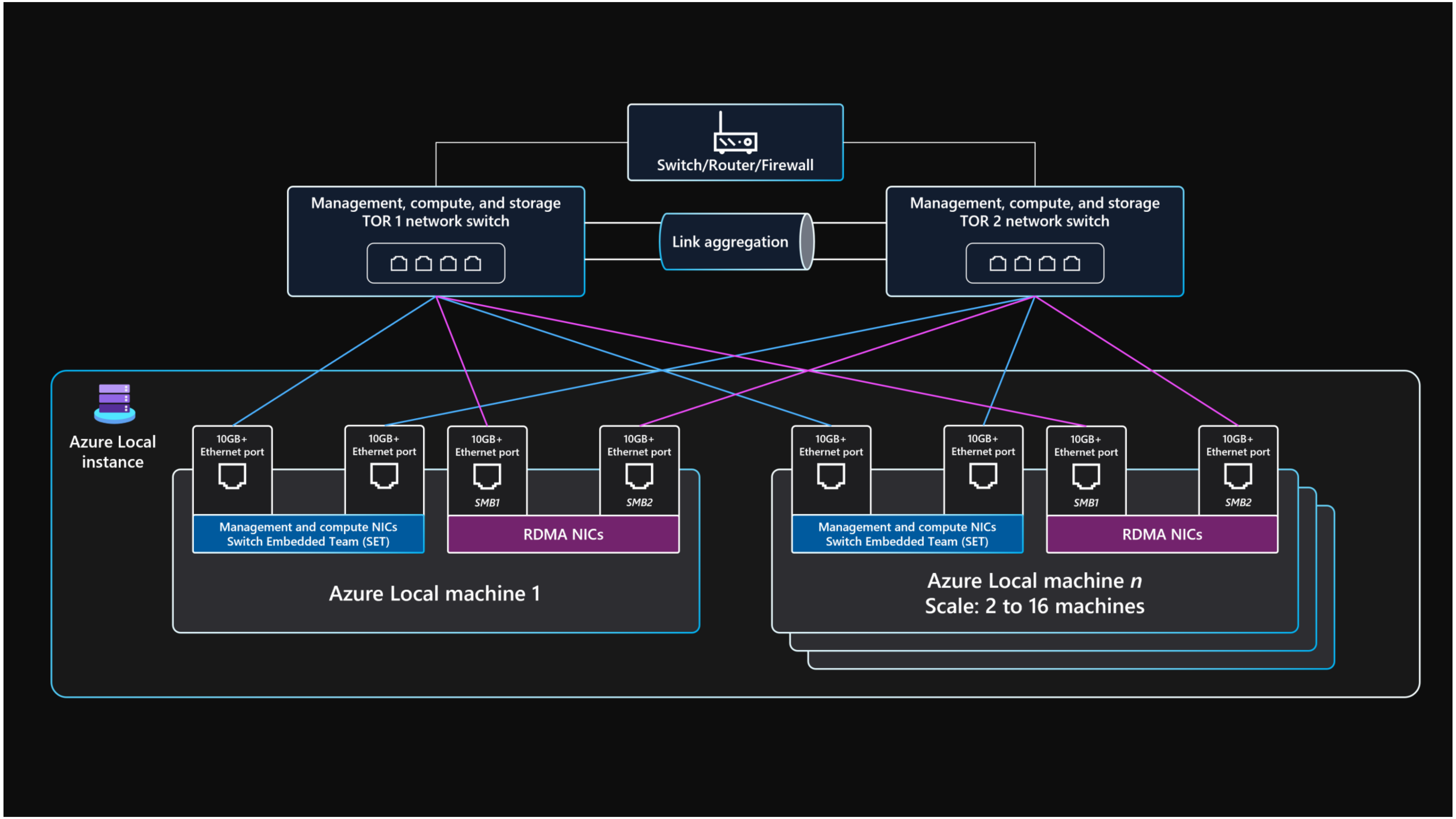
Network Architecture

- **Traffic Separation:** Dedicated RDMA NICs for storage; converged NICs for management/compute.
- **Dual ToR:** Redundant top-of-rack switches; MLAG interconnect for availability.
- **VLANs:** Non-routable storage VLANs (e.g., 711/712); converged VLANs for mgmt/compute.
- **Bandwidth:** Storage NICs ≥10 Gbps (25+ recommended); mgmt/compute ≥1 Gbps (10 recommended).
- **SMB Multichannel:** Provides storage link resiliency and throughput aggregation.

Physical Network Topology

- **Per Node NICs:** 4 ports total: 2 RDMA for storage, 2 converged for mgmt/compute.
- **Storage Paths:** One link per ToR for path redundancy and prioritized SMB Direct traffic classes.
- **Mgmt/Compute Paths:** SET teaming; one link per ToR; north-south traffic routed via ToR to external networks.
- **External Egress:** ToR connects to firewall/router; enforce required public endpoints.
- **Switch Firmware:** Keep ToR firmware current; support IEEE requirements for RDMA/PFC/VLAN/LLDP.

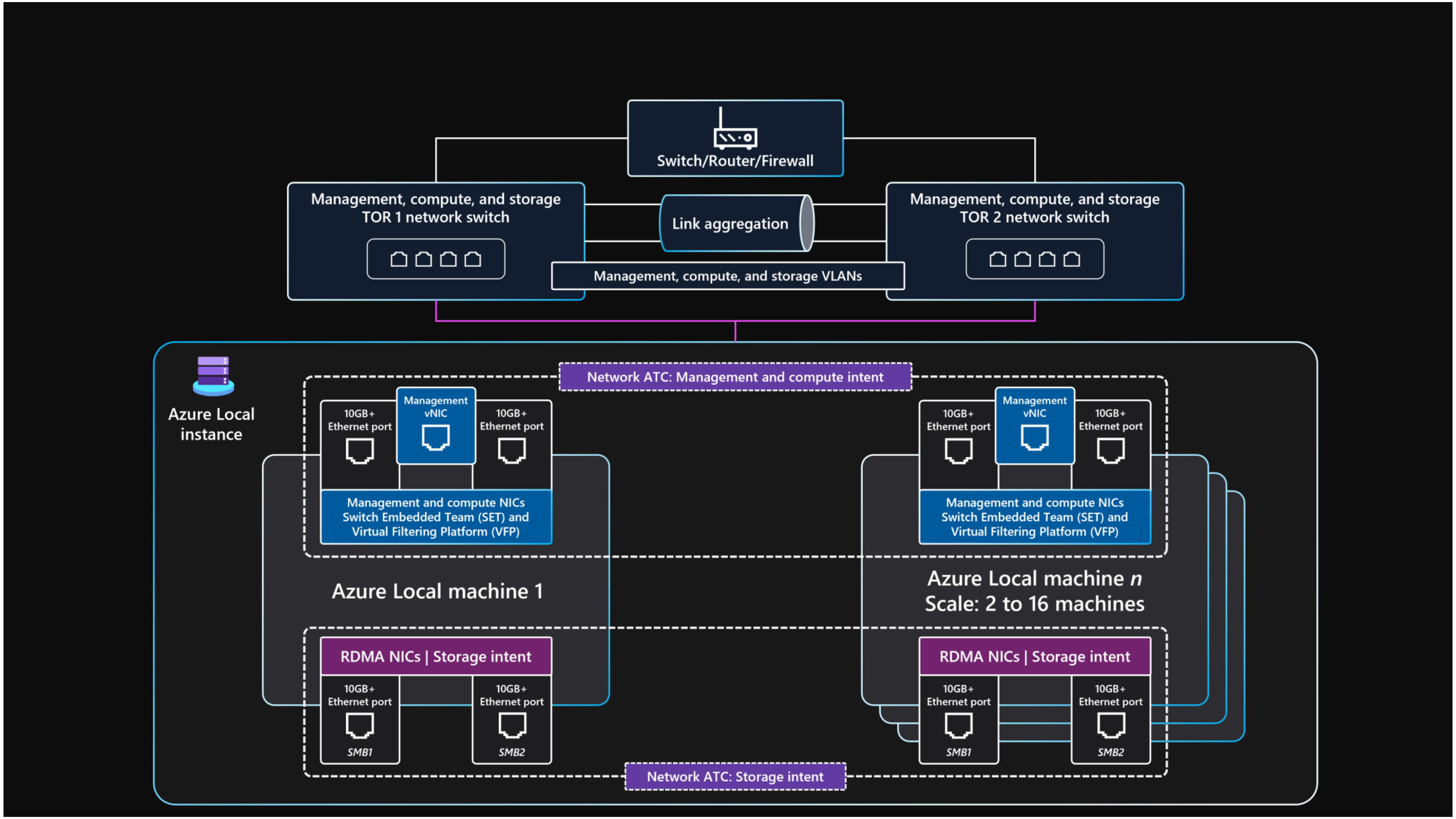
Azure Local physical network topology



Logical Network Topology & ATC

- **Network Intents:** Define management, compute, storage intents for NIC mapping.
- **Virtual Switch:** Hyper-V vSwitch with SET for converged traffic; storage NICs bypass vSwitch.
- **Network ATC:** Automates intent-based configuration; reduces misconfigurations and drift.
- **VLAN/MTU/PFC:** ATC applies validated parameters for each intent.
- **External Reachability:** Mgmt intent must reach all logical networks hosting workloads.

Azure Local logical network topology



Switch & NIC Requirements

- **RDMA: RoCE v2** or **iWARP**; **PFC** for storage QoS.
- **Standards:** Support **802.1Qbb** (PFC), **802.1Q** (VLAN), **802.1AB** (LLDP).
- **NIC Speeds:** Storage 25+ Gbps recommended; mgmt/compute 10 Gbps recommended.
- **Inter-Switch: MLAG** for redundancy; verify vendor compatibility.
- **Approved Models:** Choose Azure Local-certified switch models; validate with OEM lists.

Outbound Connectivity (Control Plane)

- **Requirement:** Outbound access to required Azure endpoints for Arc, Monitor, Policy, Update Manager, Backup/ASR.
- **Restrictions:** No TLS inspection; Private Link and ExpressRoute not supported for control-plane endpoints.
- **Arc Gateway:** Optionally reduce egress endpoint count; simplifies firewall rules.
- **Failure Modes:** If offline >30 days, cluster enters Out of Policy; Resource Bridge offline >45 days impacts keys.
- **Proxy Use:** Meet proxy requirements; ensure transparent TLS.

Azure Monitor & Insights (Cluster Observability)

- **Insights for Azure Local:** Prebuilt workbooks, metrics, health views.
- **DCR:** Data Collection Rules for performance counters and event channels.
- **Near Real-Time:** Charts, aggregation, filters; alerts for resource health.
- **Scale:** Monitor multiple clusters in a unified Azure experience.
- **Customization:** Extend with Log Analytics queries, dashboards, and alert rules.

Update Manager (Infrastructure Lifecycle)

- **Cadence:** Monthly cumulative + semi-annual feature updates (YY04/YY10).
- **Extensions/Agents:** Lifecycle updates for agents and extensions.
- **OEM Packages:** Solution Builder Extensions (SBE) of Premier Solution integrate with Update Manager
- **Compliance:** Centralized view across edge locations; track update posture.

Update Manager (Guest OS Patching)

- **Maintenance Configs:** Define schedule, reboot policy, scope (dynamic/static lists).
- **Policy Integration:** Enforce guest configuration via Azure Policy.
- **Unified Workflow:** Same portal experience for infra and guest OS.
- **Reporting:** Patch compliance across Windows/Linux VMs.

Security Foundations (Secure-by-Default)

- **Validated Hardware: TPM**, **UEFI**, Secure Boot with **VBS**.
- **OS Protections:** Application Control, Credential Guard, BitLocker for OS and S2D volumes.
- **Network Security:** SMB encryption; signed traffic; prevention of relay attacks.
- **Drift Control:** Continuous enforcement of secure baseline across nodes.
- **RBAC:** Built-in Azure Local roles for platform admins vs workload operators.

Security Operations (Defender + SIEM)

- **Defender for Cloud:** Posture management, threat detection, alerts, hardening guidance.
- **Syslog Forwarding:** Export security events for SIEM retention and analytics.
- **Compliance:** Evaluate industry standards and corporate baselines via Azure Policy.
- **Threat Analytics:** Detect AD DS threats; integrate Advanced Threat Analytics where applicable.
- **Incident Response:** Centralize triage in Azure; runbooks for remediation at edge sites.

Policy & Governance (Arc-Enabled)

- **Policy Definitions:** Evaluate resource properties; enforce guest configuration.
- **Drift Control:** Automatically reapply secure baseline; prevent config regression.
- **Scope:** Hosts, VMs, AKS; unify through Arc.
- **Activation:** Portal or IaC; target custom location resources for Azure Local.
- **Compliance Views:** Track non-compliant items; integrate with Update Manager where relevant.

Backup & DR (Azure Backup + ASR)

- **Azure Backup:** Protect Azure Local VMs; store in Backup service; policy-driven retention.
- **ASR:** Orchestrate failover to Azure for on-prem VMs; test DR without impacting production.
- **BCDR Strategy:** Daily/weekly/monthly backups; documented rotation; auditable processes.
- **Runbooks:** Define recovery, rollback, validation steps; test bi-annually.
- **Cost Trade-Offs:** Balance RPO/RTO vs storage/network egress spend.

Cost Optimization (Levers & Patterns)

- **Licensing:** Per-core monthly subscription; leverage Azure Hybrid Benefit for Windows Server Datacenter.
- **Hardware Choices:** Right-size nodes and drive tiers; avoid over-provisioning.
- **Switchless Option:** Consider storage switchless architecture to reduce switch spend.
- **Ops Efficiency:** Automate patching, monitoring; consolidate tooling in Azure.
- **Personnel Time:** Standardize with IaC; reduce manual steps; embrace GitOps for AKS.

Operational Excellence (Provisioning & Automation)

- **Portal-Driven Deployment:** Wizard for Azure Local instance creation; repeatable pattern.
- **IaC:** ARM/Bicep templates for instance and workloads (VMs/AKS/AVD).
- **CLI/PowerShell:** Local or remote operations; Arc-enabled commands for VMs and AKS.
- **Azure Automation:** Extension-based hybrid runbook workers; orchestrate maintenance and config drift remediation.
- **GitOps (AKS):** Declarative cluster/app state; policy-compliant deployments.

Performance Efficiency (Optimization Areas)

- **Storage:** Tune cache, block size, resiliency mode to workload IO profile.
- **Network:** Ensure RDMA correctness; size for burst traffic; verify PFC.
- **Compute:** Plan GPU for ML/inference; right-size vCPU/memory per VM/Pod.
- **Capacity Planning:** Model growth; watch hot spots; rebalance volumes/nodes.
- **Benchmarking:** Regular performance tests; track deviations in Monitor/Insights.

Failure Mode Analysis (FMA) — Typical Risks

- **Cluster Outage:** Power/network/hardware/software failures; mitigate with multi-instance workloads + data replication.
- **Single Node Failure:** Restart workloads on remaining nodes; recommend ≥3 nodes + three-way mirroring.
- **VM/AKS Misconfiguration:** Roll back changes; redeploy VM; reduce via Policy + IaC.
- **Azure Connectivity Loss:** Control plane degrades; workloads continue; restore within 30 days to avoid out-of-policy.

IP Addressing & Network Segmentation

- **IP Requirements:** Increase with node count; e.g., 11+ IPs for 2-node storage-switched setup.
- **Micro-Segmentation:** SDN with additional ranges for workloads
- **AKS Requirements:** Follow AKS on Azure Local network specifications.
- **Storage VLANs:** Non-routable; no default gateway on storage NICs.
- **North-South Connectivity:** Mgmt/compute via converged SET to ToR uplinks.

Arc-Enabled VM Operations

- **Provisioning:** Use Portal/CLI/ARM/Bicep targeting custom location.
- **Extensions:** Install Monitoring and Security extensions for governance.
- **Policy:** Enforce guest configuration; audit & remediate drift.
- **Lifecycle:** Patching, backup, DR; integrate with Update Manager and ASR.
- **Automation:** Scripted operations via Azure CLI/PowerShell; runbooks for repeatability.

AKS on Azure Local Operations

- **Cluster Provisioning:** Declarative creation via Azure CLI; target custom location.
- **Updates:** Use Arc extension for Kubernetes upgrades.
- **GitOps:** Enforce desired state for deployments/configs.
- **Extensions:** Container Apps, Functions, Event Grid, KEDA for scaling.
- **Observability:** Integrate Monitor/Log Analytics; set SLOs per service.

Secrets & Key Management (Key Vault)

- **Secrets:** API keys, passwords, certificates, BitLocker recovery keys.
- **Integration:** Use Key Vault in Arc-enabled workloads; rotate keys regularly.
- **Access Controls:** Apply RBAC + Policies; audit usage.
- **Compliance:** Centralize secret handling; meet regulatory requirements.
- **Automation:** Use managed identities for secret retrieval.

Switchless Storage Architecture (Cost-Focused)

- **Applicability:** 2-4 nodes; removes dedicated storage switches.
- **Trade-Offs:** Lower cost; reduced maximum scale; different network patterns.
- **When to Use:** Sites with modest capacity/resiliency needs; reuse existing mgmt/compute switches.
- **Design Considerations:** Ensure bandwidth/latency still meet workload needs.
- **Migration Path:** Maintain the ability to upgrade to storage-switched design later.

Identity & Access (Least Privilege)

- **Admin Scope:** Restrict Azure Local administrators to trusted few.
- **Monitoring:** Alert on group membership changes; audit privileged actions.
- **Role Separation:** Platform Admin vs VM Contributor/Reader roles.
- **Credential Hygiene:** Rotate local admin credentials; enforce MFA.
- **SIEM:** Centralize identity event logs for detection.

Deployment Workflow (End-to-End)

- **1) Requirements:** Gather workload scale, RTO/RPO, non-functional constraints.
- **2) Sizing:** Use Azure Local Sizing Tool; pick OEM SKUs; validate node counts.
- **3) Networking:** Deploy dual ToR, configure VLAN/MTU/PFC; map intents.
- **4) Hardware:** Rack/stack; cable; baseline firmware/driver levels.
- **5) Azure Local:** Arc onboarding; cloud deployment; extensions installed.
- **6) Workloads:** Deploy VMs, AKS, AVD targeting custom location.
- **7) Operations:** Monitor, Policy, Update, Backup, ASR; enforce runbooks.

Performance Assessment (Methodology)

- **Define Metrics:** Latency, throughput, IOPS, SLA thresholds.
- **Synthetic vs Real:** Use DiskSpd for synthetic; VMFleet for realistic mixed IO.
- **Instrumentation:** Monitor/Insights dashboards; alerts on thresholds.
- **Iterate:** Tune storage/network; retest; document baselines.
- **Capacity Forecasting:** Use trends to anticipate upgrades.

Risk Register (Top Items)

- **Outbound Egress Misconfig:** Control plane breakage; fix with Arc Gateway/TLS allowances.
- **PFC Misconfiguration:** RDMA instability; verify 802.1Qbb end-to-end.
- **Insufficient Headroom:** No N+1/N+2; outages during updates; re-size cluster.
- **Unvalidated Firmware:** NIC/switch bugs; enforce OEM update cadence.
- **Policy Gaps:** Drift from secure baseline; remediate via Policy/Update Manager.

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